MIL-H-0037210A (DSA-DM)
20 August 1976
Used in lieu of
MIL-H-37210
20 October 1975

MILITARY SPECIFICATION

HEMORRHOIDAL SUPPOSITORIES

This limited coordination military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-H-37210. It is subjected to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers Hemorrhoidal Suppositories (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be emollient, rectal suppositories which melt at body temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

3.1.1 Formula. Each suppository shall be formulated to contain the following:

Bismuth Subgallate	2.25%
Bismuth Resorcin Compound	1.75%
Nicaraguan Balsam	3.0 %
Zinc Oxide	11.0 %

and other inactive ingredients as required to make the suppository.

3.1.1.1 Batch formulation sheet. The batch formulation sheet (raw material ingredient weighing specifications) for each lot shall be reviewed to determine conformance with the formulation specified in 3.1.1 of this document.

3.1.2 Uniformity of mixture. Uniformity of mixture shall show little or no unmixed color or uncolored areas when cut longitudinally.

3.1.3 Softening point. Softening point shall be 31° to 37.5° C, when tested as specified in 4.5.1.

3.1.4 Ignition residue. Ignition residue shall be 21.4 to 24.4 percent, when tested as specified in 4.5.2.

3.1.5 Weight variation. Twenty (20) individual suppositories, selected at random, shall each weigh not less than 2.3 Grams and not more than 2.8 Grams.

3.1.6 Zinc oxide content. Shall contain not less than 9.9 percent and not more than 12.1 percent per suppository, when assayed as specified in 4.5.3.

3.1.7 Total bismuth salts (as Bi_2O_3) content. The total bismuth salt content (as Bi_2O_3) shall be not less than 1.9 percent and not more than 2.4 percent per suppository, when assayed as specified in 4.5.4.

3.2 Ingredients.

3.2.1 Bismuth Subgallate. The Bismuth Subgallate used in the manufacture of the suppositories shall be in accordance with the tests, standards, and requirements of the NF XI, which was official from 1 October 1960, to 31 August 1965.

3.2.2 Bismuth Resorcin Compound. The Bismuth Resorcin Compound shall consist of equal quantities of Bismuth Oxide, Anhydrous, Dense, and Resorcinol, USP, thoroughly mixed and milled to a fine powder, and shall comply with the following:

3.2.2.1 Bismuth Resorcin Compound.

3.2.2.1.1 Identification. Shall yield qualitative reactions for bismuth and resorcin as described in 4.5.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

3.2.2.1.2 Mesh. Ninety-eight percent minimum of a 25 gram sample shall pass through a No. 80 sieve.

3.2.2.1.3 Microscopic Examination (100X). Shall show uniformly mixed, very fine, broken crystals with no sharp edges when a thin dry mount is examined.

3.2.2.1.4 Assay. Shall assay to contain 47.5 to 52.5 percent Bismuth Oxide when assayed as specified in 4.5.5.2.

3.2.2.2 Bismuth Oxide, Anhydrous, Dense. The Bismuth Oxide, Anhydrous, Dense, used in the Bismuth Resorcin Compound shall be in accordance with the following:

3.2.2.2.1 Identification. Add a few ml of concentrated HNO_3 to about 0.5 gram of the Bismuth Oxide. Dilute the solution with water. A white precipitate colored brown by hydrogen sulfide shows the presence of bismuth.

3.2.2.2.2 Solution. A quantity of 0.5 gram of the powder shall be completely soluble in a few ml of concentrated HNO_3.

3.2.2.2.3 Arsenic. Shall conform to the USP test as applied to Bismuth Subcarbonate.

3.2.2.2.4 Assay. Shall be not less than 98% Bismuth Oxide when assayed as specified in 4.5.5.3.

3.2.3 Nicaraguan Balsam. The Nicaraguan Balsam shall consist of Peruvian Balsam, and Benzyl Benzoate, USP, so that the final mixture shall be in accordance with the following:

3.2.3.1 Specific gravity. Shall be 1.11 to 1.16 when determined in a pycnometer at 25°C.

3.2.3.2 Cinnamein saponification value. Shall be 225 to 275, when determined using the NF method as described under Peruvian Balsam (NF XIII).

3.2.3.3 Peruvian Balsam. The Peruvian Balsam used in the mixture shall be in accordance with the tests, standards, and requirements of the NF XIII, which was official from 1 September 1970 to 30 June 1975.

3.2.4 Other ingredients. All other ingredients entering into the finished suppository shall be of USP or NF quality or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.3 Expiration dating period. The expiration dating period shall be not less than 36 months.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

3.3.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.4 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintianed by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch or control is that single, uniform, and homogeneous quantity of suppositories from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with the procedures set forth in MIL-STD-105 with an inspection level of S-2 and an acceptable quality level (AQL) of 1.0 percent defective for major defects.

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

Table I. Criteria for raw material and end item testing

Characteristic	Requirement Paragraph	Inspection level 1/	Minimum Sample
Uniformity of mixture	3.1.2	S-2	5
Softening point	3.1.3	S-2	4
Ignition residue	3.1.4	S-2	2
Weight variation	3.1.6	S-2	20
Zinc oxide content	3.1.8	S-2	3
Total bismuth salts content	3.1.9	S-2	3
Ingredients 2/			
Bismuth Subgallate	3.2.1		
Bismuth Resorcin Compound	3.2.2.1		
Bismuth Oxide, Anhydrous Dense	3.2.2.2		
Nicaraguan Balsam	3.2.3		

1/ Where possible, the same shall be used for two or more tests.

2/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The Hemorrhoidal Suppositories shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

Table II. Classification of Defects

Categories	Defects 1/
Major	
101	Suppository not free of foreign matter.
102	Suppository not homogeneous (when surface is viewed and the suppository is subsequently cut longitudinally.)
103	Suppository not free of breaks, pits, or other irregularities in appearance.

1/ Inspection is not restricted to the classified possible defects listed above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

4.5 Tests.

4.5.1 Softening point.

Apparatus.

Brass plate having a thickness of 0.11 inch; with four (4) holes each having a diameter of 0.375 inch. Steel balls 0.250 inch diameter; weight 1.026 to 1.030 Gram.

Water bath. Approximately 600 ml of water in a 1000 ml beaker.

Thermometer. ASTM 36C, 45 mm, 1 mm, calibrated from - 2°C to 68°C in 0.2°C intervals.

Procedure.

Insert a suppository into each hole of the apparatus and cut off excess material so that the suppository mass just fills the hole. Make certain the entire diameter of each hole in the plate is filled with material. Place steel ball on the center of the mass filling the hole and suspend in the water bath at room temperature, placing the plate approximately midway in the water. The top of the bulb of the thermometer shall be near the top of the plate. Mechanically agitate or stir the water bath while heating. Gradually raise the temperature (temperature rise should be at 2°C per minute at softening point) of the water bath unitl each ball drops through each hole in the plate. Record the temperature for each drop. These temperatures are recorded as the softening point.

4.5.2 Ignition residue.

Weigh, accurately, about 0.7 gram sample (about 1/4 suppository), and cautiously ignite in a porcelain crucible until all carbon is burned off. Cool, add a few drops of HNO_3, evaporate to dryness on the steam bath, and re-ignite to constant weight. Cool in a desiccator and weigh.

4.5.3 Zinc oxide content.

Place 3 suppositories accurately weighed, into a 250 ml separatory funnel. Add 125 ml of ether, 3 ml conc. HCl and 10 ml of purified water. Shake until the suppositories are uniformly dispersed, then shake for an additional 5-minute period. Allow the two layers to separate and draw off the aqueous layer through a No. 597, 9 cm, S and S filter paper or equivalent, into a 200 ml volumetric flask. Repeat this extraction procedure 3 additional times, using 3 ml of conc. HCl and 10 ml purified water each time; then wash the filter paper with small portions of purified water, collecting the washings in the same 200 ml volumetric flask, until the filter is free from acid.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

Dilute the combined acid filtrates to volume with purified water and mix well. Pipet 50 ml of this solution (retain remainder for Bismuth assay) into a 250 ml Erlenmeyer flask. Add 50 ml purified water and 5 Gram NH_4Cl, warm on a steam bath or hot plate to a temperature of about 90°C. Titrate, while hot, with 0.05M Pottasium Ferrocyanide solution using 2% aqueous Uranyl Acetate solution as an external indicator. Run a blank determination on 100 ml of purified water containing 5 Gram NH_4Cl and 3 ml HCl.

Calculation:

$$\frac{(A-B) \times C \times 4 \times 100}{SW}$$ = percent of Zinc Oxide per suppository.

Where

A = ml of M/20 Potassium Ferrocyanide consumed in titration of the sample.

B = ml of M/20 Potassium Ferrocyanide consumed in titration of the blank.

C = Zinc Oxide equivalence factor.

SW = Sample weight in Grams.

Note: The Zinc Oxide equivalence factor is defined as the grams of ZnO equivalent to 1 ml of M/20 Potassium Ferrocyanide solution. It is determined by titration of a 200 mg sample of Zinc Oxide, USP, dissolved in 6 ml conc. HCl and 100 ml purified water containing 5 Gram of NH_4Cl. Prior to weighing the 200 mg sample of Zinc Oxide, USP, the Zinc Oxide should be ignited for a few minutes to drive off any absorbed water.

4.5.4 Total bismuth salts content.

Transfer 100 ml of the filtrate retained in the zinc assay into a 250 ml beaker, then pass H_2S through the solution until all the Bismuth has been precipitated as Bismuth Sulfide. Filter the solution, quantitatively, through a tared medium porosity sintered glass filter. Wash the precipitate with purified water and then with alcohol followed by ether. Cover the precipitate with carbon disulfide and allow to stand for 15 minutes. Remove the carbon disulfide by suctions and dry the residue for 2 hours at 100°C.

Calculation:

$$\frac{A \times 0.906 \times 2 \times 100}{SW}$$ = percent Bi_2O_3 per suppository.

Where A = mg of Bi_2S_3 . SW = sample weight in Grams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

4.5.5 Ingredients.

4.5.5.1 Identification (Bismuth Resorcin Compound).

Shake about 0.5 gram of the sample with 50 ml of distilled water. Filter and add 3 drops of ferric chloride T.S. to 10 ml of the filtrate; a bluish-violet color, which becomes brownish-yellow on the addition of ammonia T.S. shall show presence of Resorcinol. Wash the residue on the filter paper with a little distilled water. Transfer the residue to a test tube and dissolve in 1 or 2 ml of concentrated HNO_3 . Dilute the solution with water. A white precipitate, which is colored by H_2S shows presence of Bismuth.

4.5.5.2 Assay (Bismuth Oxide in Bismuth Resorcin Compound).

Accurately weigh 0.5 to 1 gram of sample in a tared porcelain crucible. Ignite at low temperature without melting. Cool. Take up the residue in 2 or 3 ml of HNO_3 , added dropwise, evaporate to dryness on the steam bath, ignite to Bi_2O_3, cool, and weigh.

4.5.5.3 Assay (Bismuth Oxide, Anhydrous).

Dry about 1 gram of Bismuth Oxide over sulfuric acid for 18 hours, weigh accurately in a tared porcelain crucible, and ignite to constant weight.

4.6 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of produce shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (see 6.3) and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One box containing twelve individually packaged suppositories, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. One suppository shall be packaged in an immediate container considered acceptable by the Food and Drug Administration (FDA) and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Twelve individually packaged suppositories shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. The box may be overwrapped in clear material (cellophane) which shall be secured to prevent accidental opening.

5.2.2.3 Intermediate package. Twelve unit packages shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification. As an alternate, the intermediate package may be a snug, sealed, clear plastic shrink wrap.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing.

5.3.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the appendix of the box specification.

5.3.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the box specification.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial (export when applicable) containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear label information in accordance with commercial practice.

5.4.1.2 Unit packages. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

(c) the expiration date.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in MIL-STD-129. Type I Shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

NOTE: When wrapping (cellophane) is used as the intermediate package, the required information may be printed on a label, inserted within the wrapper in such a manner as to permit ready identification.

5.4.3 Exterior container. Exterior container shall be marked as specified in MIL-STD-129. Type I Shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15°-30°C (59°-86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

6.2 This specification covers the following item(s)

National Stock No.	Item Identification
6505-01-023-3319	HEMORRHOIDAL SUPOSITORIES, 12s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1764

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. *DoD contractors, government activities, or manufacturers/ vendors* who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
HEMORRHOIDAL SUPPOSITORIES MIL-H-0037210A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

☆ GPO: 1974— 603-680/ 6126 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T10:12Z
Check the source to verify that this is the current version before use.